

Smart Fish Tank Water Quality Monitoring System

Abstract

Aquarium fish are highly sensitive to changes in water quality, making continuous monitoring essential for maintaining a healthy environment. This project proposes an IoT-based Smart Fish Tank Water Quality Monitoring System that uses an ESP32 microcontroller and multiple sensors to measure important water parameters such as pH, temperature, turbidity, and Total Dissolved Solids (TDS). The system collects real-time data from the fish tank and sends it to an Android mobile application, where the readings are displayed and analyzed to determine whether the water conditions are suitable for fish. If any parameter exceeds the safe range, the application generates an alert notification and provides recommendations on the necessary actions, such as changing the water or adjusting environmental conditions. This system helps aquarium owners and fish farmers monitor water quality continuously without manual testing, enabling early detection of harmful conditions. By providing real-time monitoring and guidance through a user-friendly mobile interface, the proposed solution aims to improve fish health, reduce fish mortality, and simplify aquarium maintenance in a cost-effective and efficient manner.

Methodology

This project develops an IoT-based fish tank water quality monitoring system that continuously measures important water parameters to maintain suitable conditions for fish. The system uses an ESP32 microcontroller connected to four sensors: a pH sensor, temperature sensor, turbidity sensor, and Total Dissolved Solids (TDS) sensor. The system is powered using a 12V power pack, and the voltage is stepped down to 5V using a buck converter to safely supply power to the ESP32 and sensors. The ESP32 collects the sensor readings and transmits the data through Wi-Fi communication to a Firebase Realtime Database, where the data is stored and synchronized with an Android mobile application. The mobile app retrieves the real-time sensor values, displays them to the user, and analyzes whether the values are within the safe range for fish. If any parameter exceeds the acceptable range, the application generates an alert and suggests corrective actions.

Before deploying the system, sensor calibration is performed to ensure measurement accuracy. The pH sensor is calibrated using a pH meter, the temperature sensor is calibrated using a thermometer, the turbidity sensor is calibrated using a turbidity meter, and the TDS sensor is calibrated using a TDS meter. During calibration, several real measurements are taken using these reference meters, and the corresponding voltage values from the ESP32 are recorded for each sensor. These voltage values are then compared with the actual meter readings to establish a calibration relationship between the sensor output and the real-world measurement values. This calibration process improves the accuracy of the system and ensures that the sensor readings closely match the actual water quality conditions.

Result and Analysis

The sensor calibration process improved the accuracy of the measurements. By comparing the sensor voltage values with the readings obtained from standard reference devices (pH meter, turbidity meter, and TDS meter), a proper relationship was established between the raw sensor data and actual values. After calibration, the sensor readings showed close agreement with the reference measurements, with only minor deviations observed, which

are acceptable for practical aquarium monitoring applications. The temperature sensor was not calibrated manually, as it operates using a built-in library that provides pre-calibrated and reliable temperature readings.

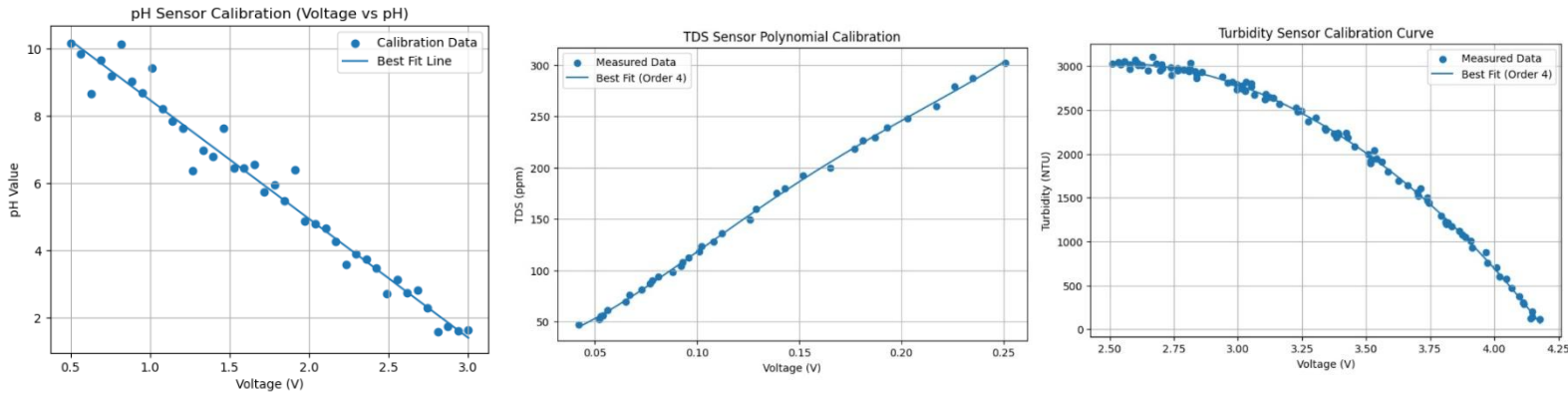


Figure 1 sensor calibrating curve

Typical safe ranges (for many aquarium fish):

pH : - 6.5 – 7.5

TDS :- 150 – 300 ppm

Turbidity :- below 25 NTU

Temperature :- 24 – 27°C

Discussion and Conclusion

The developed IoT-based fish tank monitoring system was able to effectively measure and analyze key water quality parameters, including sensors in real time. Based on testing, the system accurately identified whether the water conditions were within the typical safe ranges for aquarium fish. When any parameter moved these safe limits, the system successfully generated alerts through the mobile application and provided appropriate recommendations. This demonstrates that the system can support timely decision-making to maintain a healthy aquatic environment.

In conclusion, the proposed system provides a reliable, low-cost, and user-friendly solution for continuous water quality monitoring in fish tanks. It reduces the need for manual testing and helps prevent harmful conditions that may lead to fish stress or death. Although minor variations in sensor readings may occur, the system performs effectively for practical use, and its accuracy can be maintained through proper calibration and regular maintenance. Overall, the system is suitable for both home aquarium users and small-scale fish farming applications.